

BUSINESS ANALYTICS (23CSE452)

INDIVIDUAL CASE STUDY

What Makes a Bengaluru Restaurant Highly Rated and Popular?

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Business Analytics — Individual Case Study

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Introduction

Food delivery services and apps like Zomato and Swiggy, as well as online restaurant discovery services and apps such as Google Maps, have made it so that now virtually all restaurants conduct two business operations - the dine-in service and the delivery/ take-out service from the same kitchen. The present case study makes use of the dataset of Bengaluru restaurants, which was gathered straight from Google Maps to find out the actual difference between the highly-rated restaurant and average one and whether being popular (having high number of reviews) equals being good

1. Problem Statement and Objectives

Business Domain: The problem is related to the relationship between the information available online to users (rating, number of reviews, price, service availability) and the performance of restaurants in an Indian mega city

Problem Statement: Restaurant owners and delivery services primarily rate only one single item which combines quality of dine-in experience (dining space, ambiance and table reservation) with quality of delivery service experience (packaging, take out and online order). Therefore, two restaurants can have the same rating score for entirely different reasons and the number is not informative enough for restaurant owners to understand how to improve or what not to. Moreover, the rating does not tell anything about the popularity of the restaurant (number of reviews).

Why This Matters: Not being able to distinguish factors that influence rating and popularity, the owner may make an incorrect choice of investment - for example, spending money to change the dining area while the actual issue is with delivery service. It is a common issue among smaller and medium sized businesses that do not have a budget to conduct market research. Not being able to distinguish factors that influence rating and popularity, the owner may make an incorrect choice of investment - for example, spending money to change the dining area while the actual issue is with delivery service. A frequent problem with smaller and medium sized businesses that can't afford market research.

Objectives:

- Which of the following attributes of the restaurants - price, review volume, type of cuisine, location and service facilities (such as parking, delivery, making reservations etc.) - are statistically significant for high customer ratings?
- Whether the attributes that are used to rate customers are also used to measure popularity (in terms of the number of reviews received) or whether there are other attributes used to measure popularity and/or customer ratings?
- Develop and compare classification models that will help predict high ratings of restaurants, and then use these models to determine which attributes of the restaurants are more important in this process.

Expected Business Impact: It will provide restaurant owners with a better understanding of which services to invest into - parking, reservation or delivery service. This same analysis can be applied to delivery platforms to decide whether or not restaurants need training. If you're an entrepreneur, you'll have more information before you open your restaurant with this research.

2. Data Collection and Dataset Description

Collection method: Data was collected via the Apify Google Places Scraper from Google Maps, an open-source website with a list of restaurants. The data was filtered to obtain the information about restaurants in Bengaluru, Karnataka. Because, in the process of experimentation, direct conversion to CSV format resulted in the loss of a few columns and records in the data, the data was outputted in JSON format and not CSV format.

Source and scale: The raw dataset has 10,164 rows of data for 24 attributes.

The screenshot shows the Google Maps Scraper interface with a table of restaurant data. The table has columns for Place name, Total Score, Reviews Count, Street, City, State, Country, Website, Phone, Categories, and URL. The data is as follows:

Place name	Total Score	Reviews Count	Street	City	State	Country	Website	Phone	Categories	URL
NHB Restaurant	4.1	1688	102/1, Silver County Rd	Bengaluru	Karnataka	IN	https://www.yumyhb.com/		4 items	https://www.google.com/maps/place/NHB+Restaurant/@12.91870,77.63330,15z
KFC	4.4	1706	Bank Officers & Officials CHS, WN 197, Kanakapura...	Bengaluru	Karnataka	IN	https://restaurant.kfc.co.in/AT...	+91 80 4595 3062	5 items	https://www.google.com/maps/place/KFC/@12.91870,77.63330,15z
Muna Bhramaram Restaurant	3.9	126	Royal Meenakshi Mal, 81/3, Cave Temple Rd, opp...	Bengaluru	Karnataka	IN	https://www.munabramaram.com/	+91 76990 33001	6 items	https://www.google.com/maps/place/Muna+Bhramaram+Restaurant/@12.91870,77.63330,15z
Nagarjuna Restaurant	3.7	360	1st Floor, Bren Marcury, Sarjapur Main Rd	Bengaluru	Karnataka	IN	https://www.nagarjunarestaurant.com/	+91 91870 03330	2 items	https://www.google.com/maps/place/Nagarjuna+Restaurant/@12.91870,77.63330,15z

Important attributes collected:

- title, categoryName (cuisine) - restaurant identity and cuisine/category
- price - price range as listed (later cleaned into a numeric value)
- totalScore - the average customer rating out of 5 (referred to as "Rating" throughout this report)
- reviewsCount - number of customer reviews, used as the proxy for popularity
- imagesCount - number of customer-uploaded photos
- address, neighborhood, city, postalCode, location.lat/lng - location fields
- reviewsDistribution.oneStar ... fiveStar - a star-by-star breakdown of the review count
- Features - a free-text field listing amenities (parking, delivery, reservations, etc.), later converted into 9 separate binary columns

Identifying data, such as phone numbers and addresses, were included when publicly listed by the businesses on their own Google Map entry. No customer data was gathered – just aggregated business rating scores and number of reviews.

3. Data Preparation and Exploratory Analysis

3.1 Data Cleaning and Preprocessing

Row filtering: Restaurants that did not have any reviews or had no count of reviews were removed as they do not provide a rating signal and only noise. This took the dataset from 10,164 to 6,335 rows (3,829 removed, about 38%). No exact duplicate rows were found.

Price cleaning: Price as text string (e.g. ₹200-400) was converted to price_numeric and statistical calculation was based on the average value (midpoint) of the price. In calculating the IQR, 581 restaurants have a price that is above the upper boundary of the IQR (i.e. ₹599) and they were not dropped, as they are fine dining restaurants and not outliers. In the distribution, the numbers are skewed to the right and there are a few expensive restaurants that would skew the mean value, so the missing 1622 values were replaced with the median value of ₹300.

Target variable: High Rated = 1 if rating $\geq$ 4.0, otherwise 0. This criterion makes the problem into a binary classification task. Out of 6,335 restaurants 4,516 (71.3%) are High Rated and 1,819 (28.7%) are not. The target variable is significantly imbalanced, but the models will be created keeping this in mind.

Feature engineering: The Features section, which is free-text, has been parsed using keyword matching, creating 9 binary sections for Reservations, Parking, Delivery, Outdoor Seating, Takeout, Vegetarian Options, Wheelchair Accessible, Family Friendly, and Pet Friendly, allowing testing of individual features as opposed to one text field containing all the information.

The cleaned dataset ended up with 6,335 rows and 35 columns (up from 24).

Metric	Before	After
Rows	10,164	6,335
Columns	24	35

Missing prices	1,622	0
Duplicate rows	0	0

Table 1: Dataset summary before and after preprocessing

3.2 Exploratory Data Analysis

Each analysis below explains why it was run, before showing the result, since picking the right technique for each question matters as much as the result itself.

Rating Distribution

Why this analysis: This entire study revolves around determining the drivers behind a good rating; therefore, the first thing that must be determined is whether restaurants cluster around a high rating or are uniformly distributed; this will affect the definition of a "High Rated" restaurant.

Method: A histogram is the standard tool for looking at how one numeric variable is distributed - it groups ratings into bins and shows how many restaurants fall into each.

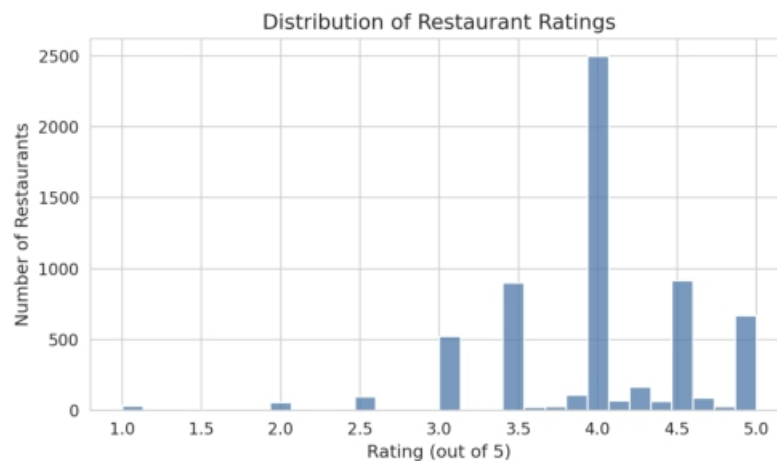


Figure 1: Distribution of restaurant ratings

What it shows: Ratings sit mostly between 3.5 and 4.5, with a mean of 3.98 and a median of exactly 4.0. Very few restaurants fall below 2.5 - the distribution is left-skewed, with bad ratings being the exception.

Business meaning: The reason why this particular criterion does not make any restaurant stand out is because almost all the restaurants are in the 3.5 to 4.5 rating range. That's why only 28.7% of restaurants were not "High Rated."

Price Distribution

Why this analysis: Price is one of the few variables a restaurant owner can control directly, so it's important to see the pricing landscape before asking whether price relates to rating.

Method: A histogram shows the overall shape of prices, while a boxplot on the same data highlights the median, the middle 50%, and outliers.

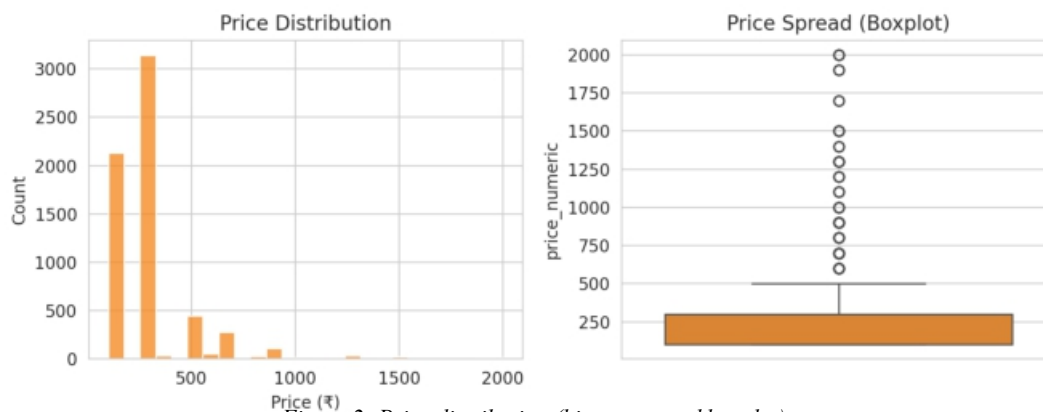


Figure 2: Price distribution (histogram and boxplot)

What it shows: Prices are heavily concentrated in the ₹100-400 range - Budget (≤ 200) and Mid-range (201-400) together make up over 80% of restaurants. The boxplot shows a long tail of outliers above ₹600, confirming a small but real fine-dining segment.

Business meaning: The restaurants of Bengaluru have a majority that falls within the budget to mid-range category. A premium-based pricing policy is fighting it out in a niche market.

Review Count Distribution

Why this analysis: Review count is used later as a proxy for popularity, so it needs to be understood on its own first: is popularity common, or do a small number of restaurants dominate it?

Method: The basic histogram would be close to unreadable in this case, due to the few restaurants that have tens of thousands of reviews and most having only a few numbers of reviews. This is what a very right-skewed data set would look like; log scaling on the x-axis makes sense in this case, as it compresses the very large figures without obstructing the shape of the main part of the data.

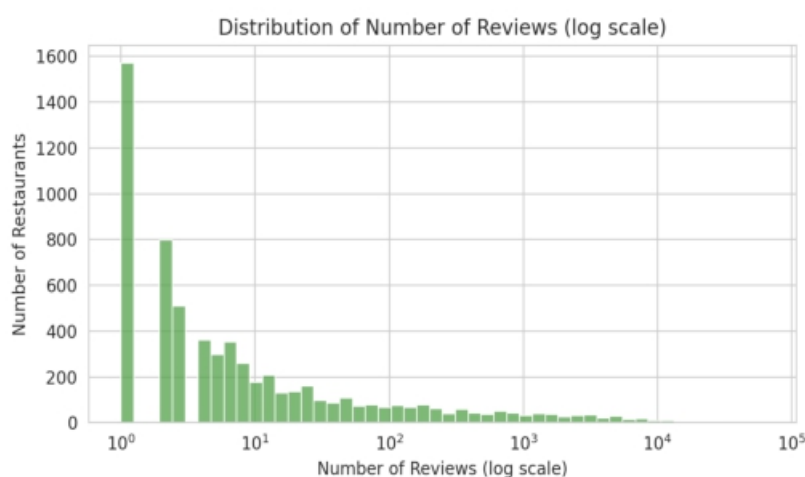


Figure 3: Distribution of number of reviews (log scale)

What it shows: On the log scale, most restaurants cluster at the low end. The median restaurant has only 4 reviews, while the mean is pulled up to about 278 by a small number of very popular places.

Business meaning: A "typical" restaurant in this case would be a small one with very few reviews – the highly reviewed chain restaurants would be an exception. The mean is misleading in this case for business reporting, whereas the median would give a truer picture.

Correlation Heatmap

Correlation can only calculate correlations between numeric values in a linear way, and Pearson correlation needs both variables to be numeric – that is why the above heatmap only contains the main numeric columns (Rating, Price, Reviews Count, Images Count). Other categorical variables (like Cuisine, Neighbourhood, or yes/no values for Delivery and Parking services) cannot go into a correlation heatmap at all – they will be considered separately below with the help of dedicated tests for categorical variables (chi-square, t-test, ANOVA).

Why this analysis: Checking correlation is the fastest way to screen the numeric variables for any obvious relationship with Rating before running more detailed tests.

Method: A correlation heatmap shows the Pearson correlation coefficient (from -1 to +1) between every pair of numeric columns, colour-coded so strong relationships stand out.

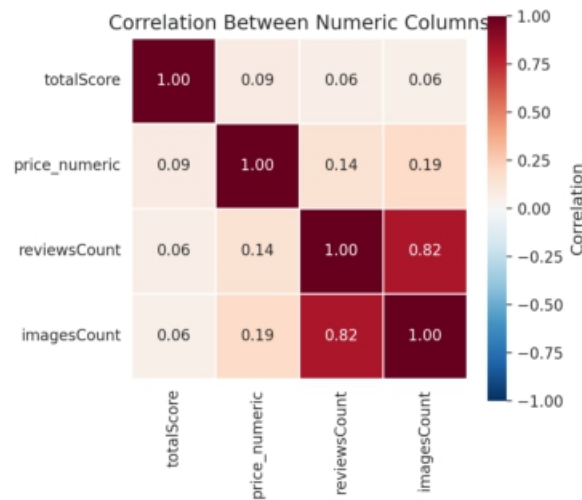


Figure 4: Correlation heatmap of core numeric columns

What it shows: Rating has only a weak relationship with Price (~0.09), Reviews Count (~0.06), and Images Count (~0.06). The strongest relationship in this heatmap is between Reviews Count and Images Count (~0.82) - not with Rating at all.

Business meaning: None of the available numeric variables can serve as good predictors for rating on its own – the answer to “what determines rating?” is definitely not singly numeric in nature.

Cuisine Analysis

Business question: Do some cuisines consistently receive better ratings than others? Cuisine is something a new restaurant owner actually chooses, so if it mattered, it would be a useful decision.

Average rating by cuisine (restricted to cuisines with at least 30 restaurants) ranges from about 4.2 (Restaurant/Fast Food) down to about 3.7 (Pizza, Chinese Thai).

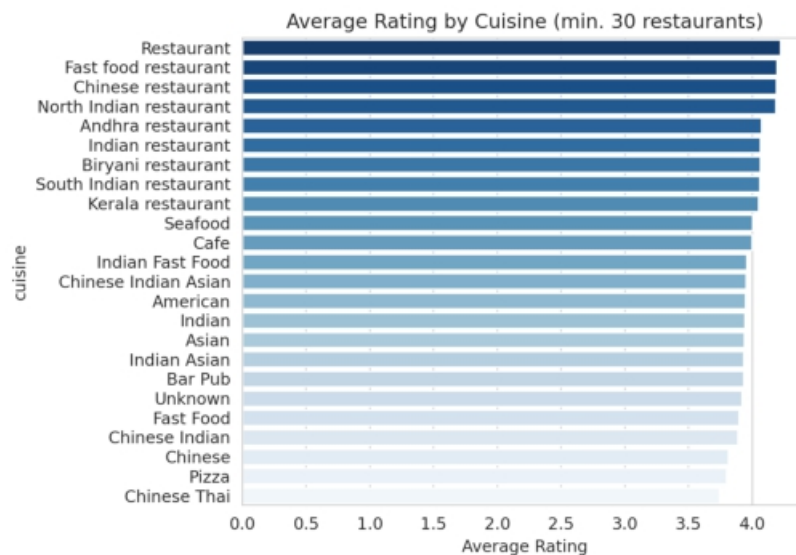


Figure 5: Average rating by cuisine (min. 30 restaurants)

Since this involves a categorical variable (cuisine) against a categorical outcome (High Rated), a Chi-Square test of independence was used rather than correlation. The test came back not statistically significant ($p = 0.44$).

Business meaning: The slight variations in the chart presented above could probably be considered as natural variations because the opening of a new restaurant does not guarantee either success or failure by the type of cuisine being offered.

Neighbourhood Analysis

Business question: Does restaurant location influence customer ratings? Location is one of the biggest fixed costs in opening a restaurant.

Among the top 15 neighbourhoods by restaurant count, average rating ranges narrowly from 3.83 (Basavanagudi/Malleshwaram) to 4.10 (HSR Layout/Marathahalli).

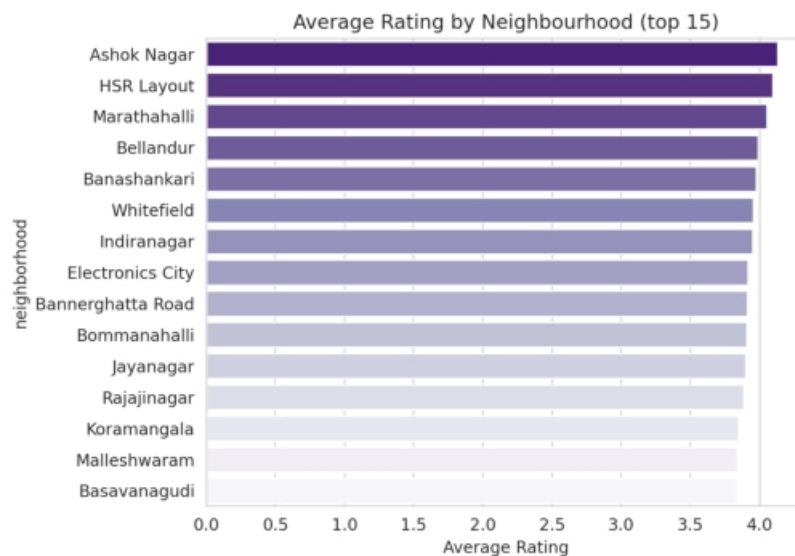


Figure 6: Average rating by neighbourhood (top 15 by restaurant count)

The same Chi-Square approach used for cuisine was applied here - also not statistically significant ($p = 0.28$).

Business meaning: As is the case with cuisine, there is only a weak and statistically inconclusive trend for neighborhoods. It could still be important to location-based traffic or rental fees, but the information presented here does not back up that assumption.

Service Analysis

What made it necessary to extract services from Features: The Features column in its raw form included all of the services, such as parking, delivery, and reservations, in a single unstructured text format that could not be analyzed through any statistical methods. Dividing it into 9 binary columns allows asking a more relevant business question: what operational services correlate with higher ratings and popularity?

Why this analysis: Since price, cuisine, and neighbourhood all turned out to be weak in predicting, service features are the next place to look.

Method: For each of six main services, an independent t-test compares the average rating of restaurants with the service against those without it - the appropriate test for comparing a continuous variable's mean between two groups.



Figure 7: Average rating with vs without each service feature

What it shows: All six t-tests came back significant. Parking showed the widest rating gap (4.21 vs 3.92), followed closely by Vegetarian Options (4.22 vs 3.96). Delivery had the smallest rating gap (4.14 vs 3.94) - but the biggest jump in review count, with a median of 127 reviews with delivery versus just 3 without it.

Business meaning: Services such as Parking do actually correlate with higher ratings, and Delivery is important primarily for review count rather than quality of rating - restaurants seeking better rating versus restaurants seeking higher number of clients will require different investment.

Popularity Analysis

It should be noted that a restaurant may attract lots of reviews, yet have low ratings, and the other way around. Here review counts are employed to serve as the indicator of popularity.c

Why this analysis: Business question: does restaurant popularity depend more on being good (rating) or on other factors like price and services? This matters because a delivery platform or investor might chase popularity assuming it means quality.

Method: A popularity score was built by log-scaling review count and rescaling it to 0-100, then split into Low / Medium / High groups. Average rating and price were compared across groups; ANOVA was used since it compares means across more than two groups.

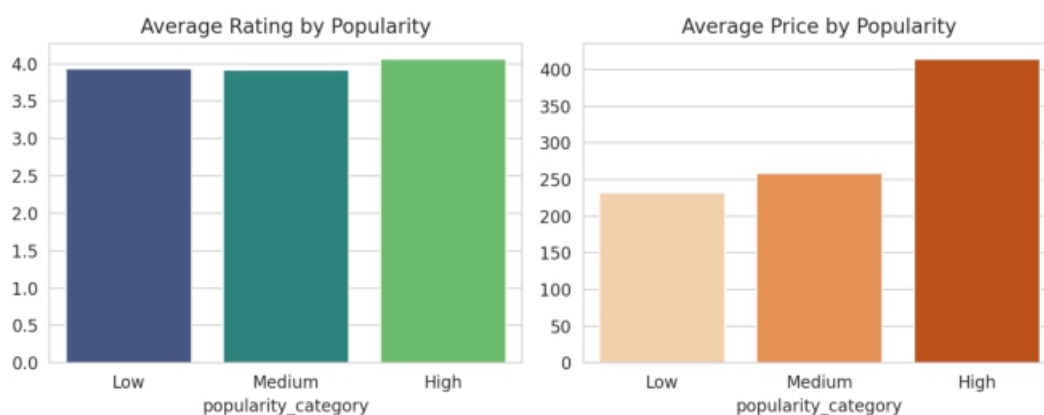


Figure 8: Average rating and average price by popularity category

What it shows: The High popularity group has only a slightly better average rating (4.07) than Low (3.94) or Medium (3.92). Average price rises much more clearly across the same groups (₹232 → ₹259 → ₹415). ANOVA confirmed price is far more strongly tied to popularity ($F = 327$) than to rating ($F = 19.82$) - over 15 times larger.

Business meaning: The popularity within this data set is far more correlated with price and services offered than with ratings, which is a helpful reminder to all those who equate “popularity” with “excellence”.

4. Analytics Method and Implementation

4.1 Re-checking the Target Variable

It is worthwhile to consider if High Rated (Rating ≥ 4.0) is an appropriate target before going about development. The business goal is actually split into two components – what makes a difference to the rating, and what makes a difference to the popularity. As the popularity attribute (reviewsCount) does not have any obvious cut-off criterion to define ‘popularity’ as a separate attribute, the question of popularity was solved through the statistical approach discussed above (Popularity Analysis, ANOVA), rather than a separate classifier. A regression model was also fitted on the actual value of the rating, but the rating distribution (Figure 1) is very skewed with a number of popular values, and a regression model would essentially be predicting the same number of values repeatedly.

4.2 Feature Selection

Features were selected, not because they were available, but because they have the potential to answer the questions of which characteristics affect rating and popularity.

The price_numeric, reviewsCount, and imagesCount (pricing and popularity signals) are all included as well as the 9 engineered service features (Reservations, Parking, Delivery, Outdoor Seating, Takeout, Vegetarian Options, Wheelchair Accessible, Family Friendly, Pet Friendly).

Excluded: totalScore (for target leakage, since High Rated is derived from this), all the columns under the heading “reviewsDistribution” (because it's another decomposition of the target, and you'll see a leakage problem there), cuisine, neighborhood (tested with Chi-Square test, and found to have no statistically significant association with the target – effect size for cuisine as per Cramer's V is almost negligible), price_category (duplicative of the numeric price_numeric feature), and all the restaurant-identifying features like title, address, phone, website, and coordinates (because they tell us about a particular restaurant, and not the restaurant characteristics in general).

4.3 Method Justification: Why Classification, Why Three Models

The business question would then be a yes/no question, and the rating was converted to the High Rated flag, giving it the classification of a binary classification problem. Instead of choosing one model ahead of time, three models were trained and compared: Logistic Regression, which is simple and has interpretable coefficients; Decision Tree, which can capture non-linear patterns of relationships and is easy to visualise; Random Forest, which is typically more stable, even though it is more difficult to interpret. Comparing all three is a way of avoiding the selection of a model that happens to fit a certain expectation.

The data was stratified sampled (75/25, 4,751 train set, 1,584 test set) to retain the same proportion of High Rated (71.3%) in both sets and numeric variables were scaled before fitting the logistic regression model. This was done as the logistic regression algorithm is sensitive to variable scaling and not the tree-based models. All variables are numeric and do not have any missing values, and the totalScore variable has been validated as not being among the input variables (no target leakage).

4.4 Model Results

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	0.713	0.713	1.000	0.832	0.551
Decision Tree	0.698	0.710	0.972	0.821	0.570
Random Forest	0.711	0.713	0.996	0.831	0.593

Table 2: Model comparison on the held-out test set

Reason for Multiple Metrics rather than only Accuracy: Accuracy metric on its own is meaningless because of imbalance between classes (71.3% High Rated) – an untrained model can reach 71% of accuracy just by predicting "High Rated" for all cases. Precision: how many among those predicted as High Rated restaurants were high rated; Recall: how many among high-rated restaurants were correctly predicted to be high rated; F1 score – compromise between them; AUC – threshold-independent measure of how good the model is in sorting restaurants by probabilities.

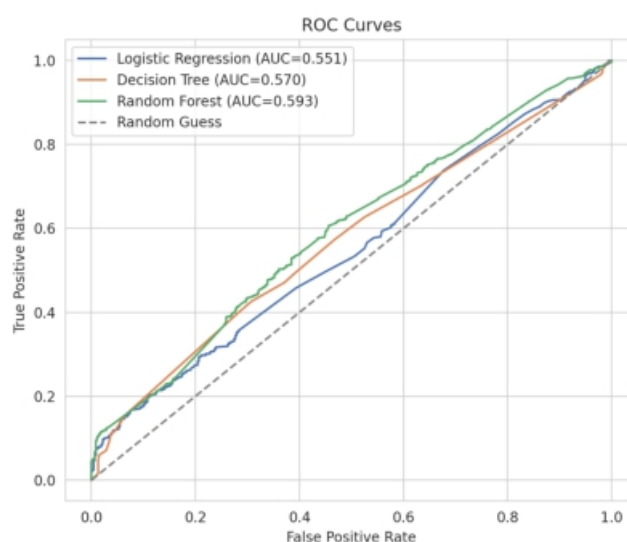


Figure 9: ROC curves for the three models

The three models all fall into the same range – 70-71% accuracy and 0.55-0.59 AUC. Random Forest had the maximum AUC and was selected to be the best model. Is there enough of the selected features? No – even by using Random Forest, the AUC is not much better than 0.593. This is a data set based on metadata of listings, not review data – food quality, taste and consistency of service are important when rating and these do not appear here and is the main contributor to the low ceiling.

A GridSearchCV was used to tune the hyperparameters, with 5-fold cross validation (optimal parameters: max_depth=8, min_samples_leaf=10, n_estimators=300), and balance class weight and threshold adjustment to handle class imbalance. Cross-validated AUC went up to 0.642, while on the test set AUC has increased only slightly, from 0.593 to roughly 0.596 – this is a very slight increase, as this result shows that the limit of predictive power of these features has already been reached.

4.5 Feature Importance Across All Three Models

All three models were tested for the importance of features, and not just Random Forest since their definition of “importance” varies: linear coefficient, the number of times the feature is used to split a Decision Tree and the average of these numbers over many Decision Trees in a Random Forest. As a fourth cross-check, the Permutation Importance test was added to the model-agnostic test that involves shuffling one feature at a time and measuring the change in AUC.

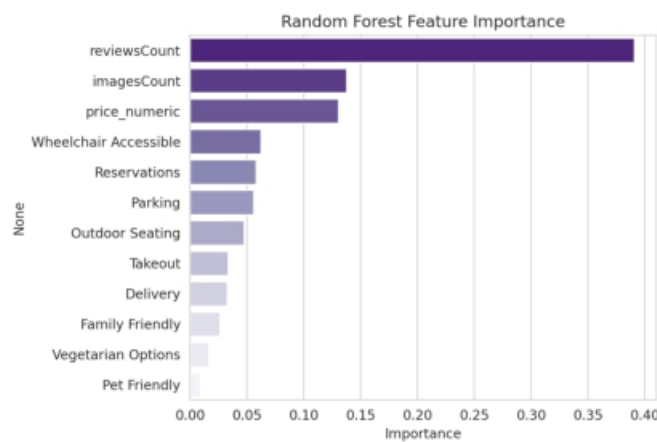


Figure 10: Random Forest feature importance ranking

Feature	LR Rank	DT Rank	RF Rank	Permutation Rank	Avg Rank
reviewsCount	8	1	1	1	2.75
Reservations	5	2	5	2	3.50
Wheelchair Accessible	2	4	4	7	4.25
imagesCount	1	6	2	8	4.25
Parking	3	8	6	4	5.25
price_numeric	12	3	3	5	5.75
Outdoor Seating	10	7	7	3	6.75
Delivery	7	5	9	9	7.50
Takeout	11	9	8	6	8.50
Family Friendly	6	10	10	11	9.25
Pet Friendly	4	11	12	10	9.25
Vegetarian Options	9	11	11	12	10.75

Table 3: Feature importance rank comparison across all four methods (rank 1 = most important)

In all three methods (Decision Tree, Random Forest, Permutation), reviewsCount is the first one to appear in the list, except for Logistic Regression. It has a negative effect (coefficient = -0.055). This is because of the high correlation of the variables (0.82) as demonstrated in Figure 4 and the multi-collinearity problem. According to the three independent methods, it can be said that reviewsCount is the only one that is the most reliable. Reservations' is the most highly ranked of all the service-related features on average (3.50) and is ranked in the top half of all four methods. It is also the most consistent finding in the category as it passed the t test successfully. The three features that are ranked the least among all methods are Vegetarian Options, Pet Friendly, and Family Friendly.

5. Comparison with State-of-the-Art Methods

Below are three recent published studies of restaurant rating and restaurant success prediction, which were identified and compared to this case study. It is not a matter of "simply better", because the datasets and experimental settings are different, but rather a matter of comparison of the method, of the dataset, of the approach to the evaluation and of the findings

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
Choudhary et al., 2021 [1]	Zomato listings, Bengaluru (17 attributes)	Linear Regression, Decision Tree, Random Forest	Accuracy	Random Forest 87%, Decision Tree 85%, Linear Reg. 24%	Method: Analyzes restaurants using the methods Linear Regression, Decision Tree and Random Forest. Data Source: Zomato Restaurant Listings in Bengaluru with 17 attributes which are geographically relevant to this study. My study evaluates Accuracy, Precision, Recall, F1 and AUC and is explicitly considering class imbalance as compared to the primary evaluation that reports accuracy (Random Forest 87%, Decision Tree 85%, Linear Regression 24%). Findings: Both studies confirm the value of a tree-based approach for restaurant analysis. Strength: Direct benchmark from bengaluru using similar ML methods. Limitation: The accuracy values reported in my study should not be compared directly with those reported in other studies because these are reported with unbalanced class distributions. My study additionally combines predictive modelling with statistical tests and multiple feature-importance techniques.
Li, Yu, Li & Gao, 2023 [2]	Online reviews, 2 major US tourism cities	Aspect-based sentiment analysis (ABSA) + ML conditional survival forest	Prediction accuracy + feature importance	Aspect-level sentiment (service, tastiness, price, location, atmosphere) outperforms overall review sentiment	Method: Based on the Aspect Based Sentiment Analysis and Machine learning Conditional Survival Forest. Customers generated online reviews from two major cities in the USA about a travel service, including detailed text information about various aspects like service, tastiness, price, location, and atmosphere. Evaluation: Prediction accuracy, Feature importance. Learning Objective: Understand the importance of aspect level information gained from customer reviews. The strength of Review text is that it supplies more information about the customer experience than the listing attributes. Limitation: The sample size and environment of this Bengaluru Google Maps study is very different from the one proposed here. Compared to this, my study is more focused on structured restaurant attributes and services provided in the restaurant and also uses Chi-Square tests, t-tests, ANOVA and popularity analysis, enabling statistical analysis along with prediction of the business factors.
Li & Hecht, 2021 [3]	Paired Google Maps + Yelp listings, multiple US metros	Statistical / comparative analysis of cross-platform rating distributions	Mean rating difference, top-10 ranking overlap	Google Maps ratings average ~0.7 stars higher than Yelp, driven by chain restaurants	Method: Conducts statistical and comparative analysis of restaurant ratings from various platforms. The data includes a set of Google Maps and Yelp listings in several US metros that were paired together. Evaluation: Mean rating differences, rather than predictive ML metrics, and top-10 ranking overlap. Finding: Platform effects are important to keep in mind when interpreting restaurant ratings, given the differences in ratings reported by Google Maps and Yelp. Suggests a helpful cross-platform context for rating distributions. Limitation: It fails to build a restaurant-rating prediction model and does not examine the operational service characteristics. The focus of my study is a single platform and Bengaluru specific restaurants along with EDA, Statistical Hypothesis testing, Popularity analysis, Classification models and Feature importance analysis.

Table 4: Comparison with three recent published studies (see References)

Note on comparison: the differences in reported metrics reported above are not to be interpreted as simply "better" - the figure of 87% reported by Choudhary et al. is a raw accuracy on a different (and likely less imbalanced) train/test split; the reported AUC is specifically because the class imbalance in this study (71/29)

makes the reported accuracy misleading. The better performance of the study of Li et al. (2023) is not only due to a better algorithm, but also because the algorithm was presented with richer input features (the review text sentiment). A fair comparison ought to take into account the algorithm used, and what it was applied to.

6. Results, Business Insights and Recommendations

6.1 Key Results

A relatively narrow range of factors - Review Volume, Reservations, Parking, and to a lesser degree, Wheelchair Accessible and price at the top end - drive rating. Price is a greater determinant of popularity than is Delivery/Takeout. Random Forest was tested and found to be the best classification model (AUC 0.593, but only marginally better at 0.596 after tuning) - indicating that the price, reviews, and service flags alone do not fully account for a highly-rated restaurant.

6.2 Business Insights

Each insight below is linked to its related analysis.

- There was a large gap between the ratings for Parking in the service t-tests (4.21 vs 3.92) and it was also a top 5 service feature in the importance ranking, two t-tests yielding the same result is a good indicator.
- The correlation between price and rating overall is very weak (Pearson ~ 0.09), but the price-category comparison found that the Fine Dining category was signaled as having a noticeably higher rating (4.28), which is confirmed by ANOVA - price matters not as a rule, but specifically at the very top end.
- There was weak correlation between the review count and rating per se but it was the top rating in 3 of 4 methods of importance, indicating that the review count is more of a supporting signal to the model than a linear predictor.
- The smallest rating gap of the six services tested was for Delivery with the largest popularity jump (median reviews 3 $\rightarrow$ 127) - real business value is in the visibility, not in the ratings.
- The differences within the bar charts for both Cuisine and Neighbourhood were small and neither were significant when tested with the Chi-Square test - unlike Parking and Price Category, these should not be considered as genuine drivers of rating.
- The popularity analysis revealed that the restaurants that are popular are not significantly better rated than the Low or Medium popularity restaurants, but that there is a significant difference in average price-popularity among these groups: In this market, the popularity is more based on price and services than on actual quality.
- The best classification models did not do significantly better than the majority-class baseline with a score of 71.3%, and had only moderate correlations with the EDA.

6.3 Recommendations

All recommendations are in the format Analysis $\rightarrow$ Result $\rightarrow$ Insight $\rightarrow$ Recommendation.

For Restaurant Owners:

The service t-tests indicated that Parking had the largest rating difference of any feature tested and was in the top ten of all the features tested in the importance comparison, so it was the highest rated feature in both tests and therefore is the strongest recommendation: Prioritise parking availability where physically possible. Volume and service quality are better investments than a general increase in price; the price correlation heatmap and price-category ANOVA showed that price alone has little impact on rating, except at the highest price range.

For Food Delivery Platforms:

The service analysis revealed that Delivery has the smallest rating gap among the six services analysed, and the largest increase in the number of reviews, suggesting that Delivery is more about visibility than quality, and so should be marketed as a reach tool rather than a quality indicator. Adoption of service was lower for budget restaurants, suggesting that the budget segment is the most obvious onboarding opportunity.

For Entrepreneurs Planning a New Restaurant:

The price distribution revealed that more than 80% of the restaurants already exist in the price band of ₹100-400, and the cuisine/neighbourhood analysis (both statistically non-significant) indicated that competing in terms of

cuisine or neighbourhood choice is not likely to be the path to victory in this segment; rather, it is execution and services such as Parking that are more likely to be the winning factors.

7. Conclusion and References

7.1 Conclusion

In this case study set, I aimed to find out what makes a restaurant good or bad and popular, and analyzed a total of 6335 restaurants from Bengaluru, which were retrieved from Google maps and cleaned up. Rating has only a weak relationship with Price and Review Count, and Service Features (particularly Parking, Reservations, and Vegetarian Options) had a small but statistically significant relationship with rating. When comparing the ANOVA results between rating and popularity, it was clear that popularity was more of a price and service availability thing, and this was confirmed by comparing it with the ratings - popularity is a different question and a different answer.

Random Forest was the best performing classification model (AUC 0.593, which improved slightly to 0.596 after tuning), and when the feature importance of the model was compared to that of three other importance methods and the previous methods of statistics, they all pointed to the same few features, which were review count and a few service features. The bottom line is that volume of reviews and a few practical services features (parking in particular) are more important than pricing strategy in their effectiveness to being considered a good restaurant. Potential future research may include sentiment analysis of reviews, rating trend over time, or distinguishing dine-in ratings from delivery ratings.

7.2 References

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